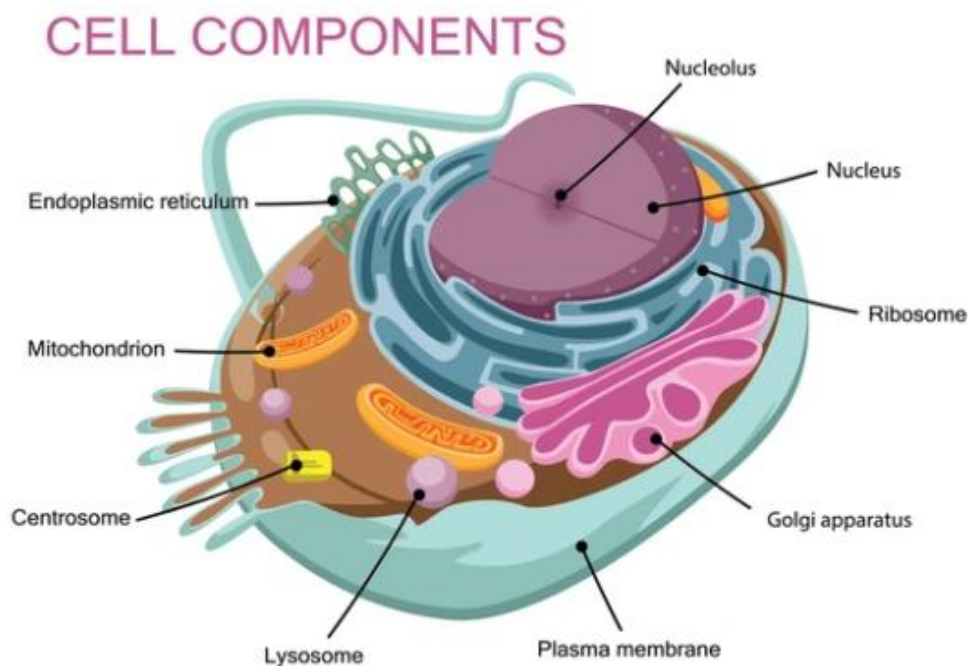


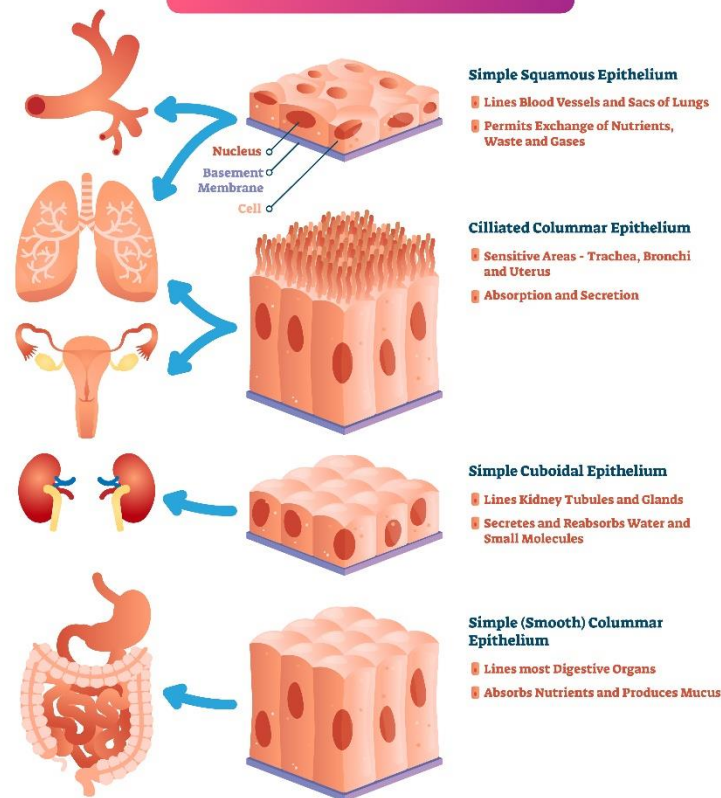
Anatomy of a Cell

A typical animal cell consists of three main parts:

- **Plasma Membrane:** A semipermeable phospholipid bilayer that acts as a barrier, regulating what enters and exits the cell.
- **Cytoplasm:** A jelly-like substance (cytosol) containing organelles, inclusions, and the cytoskeleton.
 - ❖ **Mitochondria:** Powerhouse of the cell, generating adenosine triphosphate (ATP).
 - ❖ **Endoplasmic Reticular (ER):** Network for transport/synthesis of proteins (Rough ER) and lipids (Smooth ER).
 - ❖ **Golgi Apparatus:** Modifies, sorts, and packages proteins for secretion.
 - ❖ **Lysosomes:** Contain enzymes to break down waste materials.
 - ❖ **Nucleus:** The control center containing DNA, which is housed within a nuclear envelope.



EPITHELIAL CELL



Anatomy of Tissues

Tissues are groups of cells with similar structures working together for a common function. There are four primary tissue types:

- **Epithelial Tissue:** Highly cellular tissue covering body surfaces, lining cavities, and forming glands (e.g. skin, stomach lining). It is avascular, meaning it lacks blood vessels.
- **Connective Tissue:** The most abundant tissue, providing structural support and binding other tissues together (e.g., bone, blood, fat). It consists of cells, fibers (collagen/elastic), and ground substance.
- **Muscle Tissue:** Specialized for contraction, creating movement. Types include:
 - ❖ **Skeletal:** Striated (striped), voluntary muscles attached to bones.
 - ❖ **Cardiac:** Striated (striped), involuntary muscle of the heart.
 - ❖ **Smooth:** Non-striated, involuntary muscle in organ walls.
- **Nervous Tissue:** Composed of neurons (which transmit electrical impulses) and glia (support cells).

These tissues cooperate to form organs (e.g., heart, liver) and, subsequently, organ systems (e.g., cardiovascular system) to maintain homeostasis.

